View Reviews

Paper ID
237

Paper Title
Non-Invasive Anemia Detection Using Ocular Images and Deep Learning

Reviewer #1

Questions

1. What is the nature of the article?
Research article

2. Does the paper reflects theme/scope of the conference?
Yes

3. Originality
Good

4. Innovation
Good

5. Significance
Good

6. Organisation of the manuscript
Good

7. Are the results discussed convincingly
Yes

8. References
Good

9. Language
Good

10. Quality of figures
Good

11. What is your recommendation on the paper?
Accept with major revision

12. Detailed comments about the paper (Mandatory, it will help authors to revise and improve the paper further)
Kindly refer the attachement

Files

237-review-2870.pdf (135 Kb, 2026-10-13 12:41:11 GMT+5:30)

Reviewer #2

Questions

1. What is the nature of the article?

Research article

2. Does the paper reflects theme/scope of the conference?

Yes

3. Originality

Good

Fair

4. Innovation

Good

Fair

5. Significance

Good

6. Organisation of the manuscript

Good

7. Are the results discussed convincingly

Yes

8. References

Good

9. Language

Good

10. Quality of figures

Good

Fair

11. What is your recommendation on the paper?

Accept with minor revision

12. Detailed comments about the paper (Mandatory, it will help authors to revise and improve the paper further)

The manuscript presents a relevant approach for non-invasive anemia detection using ocular images and deep learning. The topic is significant because it addresses the limitations of conventional invasive blood-based diagnosis and explores an accessible image-based screening approach.

The title is appropriate and clearly reflects the objective of the work. The abstract provides a concise overview of the proposed MobileNetV2-based feature extraction and SVM classification framework and appropriately mentions the use of SMOTE to address class imbalance while avoiding data leakage. However, the abstract would be improved by reporting the main quantitative performance results.

The literature review provides adequate background on anemia detection, ocular imaging, deep learning, transfer learning, and related machine learning approaches. The proposed methodology is clearly structured, including preprocessing, MobileNetV2 feature extraction, SVM classification, stratified splitting, and training-only application of SMOTE. However, further details regarding hyperparameter selection, feature extraction configuration, and model optimization would improve reproducibility.

The experimental results are promising, with the proposed model reporting 94.75% accuracy, 0.95 precision, 0.96 recall, 0.93 specificity, 0.95 F1-score, and 0.99 ROC-AUC. The use of multiple evaluation metrics provides a reasonably comprehensive assessment. However, the study relies on approximately 800+ ocular images from a publicly available dataset, and additional validation using larger, more diverse, and independently collected datasets would be necessary to establish generalizability.

The manuscript would also benefit from comparison with more recent state-of-the-art deep learning approaches, clearer reporting of hyperparameters and experimental settings, and external or clinical validation. The claims regarding practical screening should be stated cautiously until such validation is performed. Overall, the work is technically sound and relevant, and the identified issues can be addressed through minor revisions. Accept with minor revision is recommended.

Reviewer #3

Questions

1. What is the nature of the article?
Research article

2. Does the paper reflects theme/scope of the conference?
Yes

3. Originality
Fair

4. Innovation
Fair

5. Significance

Good

6. Organisation of the manuscript

Good

7. Are the results discussed convincingly

No

8. References

Fair

9. Language

Good

10. Quality of figures

Fair

11. What is your recommendation on the paper?

Accept with major revision

12. Detailed comments about the paper (Mandatory, it will help authors to revise and improve the paper further)

The paper extracts MobileNetV2 features from eye images and classifies them with an SVM (RBF), using SMOTE on the training split only. It reports about 95% accuracy and an AUC of 0.99 on the Eyes Defy Anemia dataset.

1. Accuracy is 94.75% in Table II but 95.75% in Tables IV and V.
2. The confusion matrix in Fig. 5 gives about 94.98%.
3. A 259-image test set at an 80:20 split implies about 1,300 images, yet the dataset is described as "800+".
4. Fig. 4 shows SVM, Logistic Regression, Random Forest, and KNN, but the text and tables compare SVM with XGBoost, which does not appear in the figure.